Formula for Maximum Gap

The formula for the maximum gap is:

$$\theta = \max\left(P(x+1 \mid x) - P(x \mid x-1)\right)$$

Where:

- $P(x+1 \mid x)$ is the conditional probability of having $x+1$ pellets given x pellets.
- $P(x \mid x-1)$ is the conditional probability of having x pellets given $x-1$ pellets.

Implementation in Python Code

In your Python code, the maximum gap is calculated in the `determine`$_{t}hreshold_{m}ax_{g}apfunction.Here'showiti$

1. Calculate Differences: The differences between consecutive conditional probabilities are calculated using np.diff(probs).

    ```
    diffs = np.diff(probs)
    ```

2. Find Maximum Drop: The index of the maximum drop (largest negative difference) is found using np.argmin(diffs).

    ```
    max_drop_index = np.argmin(diffs)
    ```

3. Set Threshold: The threshold is set slightly above the drop point by adding the difference at the maximum drop index to the probability at that index.

    ```
    threshold = probs[max_drop_index] + diffs[max_drop_index]
    ```

4. Ensure Valid Threshold: The threshold is ensured to be between 0 and 1.

    ```
    threshold = min(max(threshold, 0.0), 1.0)
    ```

Here's the relevant part of your code with comments for clarity:

```
def determine_threshold_max_gap(conditional_probabilities):
    sorted_sizes = sorted(conditional_probabilities.keys())
    probs = [conditional_probabilities[size] for size in sorted_sizes]

    # Calculate differences between consecutive probabilities
    diffs = np.diff(probs)

    if len(diffs) == 0:
```

```
        print("Not enough data to determine threshold.")
        return 0.7  # Default threshold

    # Find the index of the maximum drop
    max_drop_index = np.argmin(diffs)

    # Set threshold slightly above the drop point
    threshold = probs[max_drop_index] + diffs[max_drop_index]

    # Ensure threshold is between 0 and 1
    threshold = min(max(threshold, 0.0), 1.0)

    print(f"Determined Threshold based on Maximum Gap Method: {threshold:.4f
    return threshold
```

This function identifies the point where the change in conditional probabilities is most significant (the maximum gap) and sets a threshold based on this gap. This threshold can then be used to analyze the behavior of the mice.